

ITERATING LYNDON-FACTOR STARTS ON BINARY WORDS

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ABSTRACT. Fix $0 < 1$ and replace a binary word by the mask of the starting positions in its nonincreasing Chen–Fox–Lyndon factorization. We iterate this mask map. Factor starts are exactly the left-to-right strict record minima among the suffixes. Every leading 1 is a singleton Lyndon factor, which makes the leading-one prefix grow under every nonfixed update. It follows that 1^n is the unique recurrent state and every orbit reaches it within n steps. The bound is sharp at a unique state: the alternating word $0101 \dots$. For every target mask, we also identify the complete fibre with a set of nonincreasing chains of binary Lyndon words of prescribed lengths and express its size as a product of rectangular comparison matrices. The one-factor and all-singleton targets give the classical binary Lyndon census and $n + 1$, respectively. All static factorization and enumeration results are credited background. External release remains on hold.

1. FACTOR-START FEEDBACK AND THE CREDIT BOUNDARY

Words are ordered lexicographically over $0 < 1$, with a proper prefix smaller than the longer word. A nonempty word u is *Lyndon* if it is strictly smaller than each of its nontrivial rotations. The classical Chen–Fox–Lyndon theorem [2] gives every nonempty word a unique factorization

$$(1) \quad w = u_1 u_2 \dots u_k, \quad u_1 \geq u_2 \geq \dots \geq u_k,$$

into Lyndon words; repeated factors are written separately.

Let $1 = s_1 < s_2 < \dots < s_k \leq n$ be the factor starts in (1). Define the binary self-map $L_n : \{0, 1\}^n \rightarrow \{0, 1\}^n$ by

$$(2) \quad L_n(w)_i = 1 \iff i \in \{s_1, \dots, s_k\}.$$

The factorization is recomputed from the mask after every update.

Duval computes (1), the least suffix, and the least circular shift in linear time [3]. Lyndon arrays and forests have their own algorithmic theory [1, 4]. Those algorithms, the CFL theorem itself, the binary Lyndon census, necklace Möbius inversion, and matrix multiplication receive zero contribution credit here. Our claims concern only the repeated map (2), its sharp clock, and its inverse atlas.

A state is recurrent if it lies on a directed cycle. Its depth is the first time its orbit reaches a recurrent state.

2. FACTOR STARTS ARE SUFFIX RECORDS

We use two standard consequences of the Lyndon definition. A Lyndon word is unbordered and is strictly smaller than each of its nonempty proper suffixes. We also use the ordered-tail comparison underlying the CFL theorem: if $u_1 \geq \dots \geq u_k$ are Lyndon and $S_h = u_h \dots u_k$, then

$$(3) \quad S_1 > S_2 > \dots > S_k.$$

Here is the short reduction, including equal factors. It is enough to compare adjacent tails. Fix h , put $u = u_h$, and remove from both tails the common initial copies of u occurring in S_{h+1} . If no word remains on the right, that right-hand word is a proper prefix. Otherwise it begins with a Lyndon factor $v < u$; the classical Lyndon concatenation comparison says $uz > z$ whenever the CFL factors of z begin with $v < u$ and thereafter do not increase. Thus $S_h > S_{h+1}$ in both cases. This is a static comparison lemma, not a dynamical contribution; it is also the comparison step in the standard proof of the CFL factorization [2, 3].

Proposition 2.1 (suffix-record characterization). *For $w = w_1 \dots w_n$, position i starts a factor in (1) if and only if its suffix is a strict new minimum among the suffixes seen from the left:*

$$(4) \quad L_n(w)_i = 1 \iff w_i \dots w_n < w_j \dots w_n \text{ for every } j < i.$$

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Proof. Write $S_h = u_h u_{h+1} \cdots u_k$ for the suffix beginning at factor u_h . By (3), each later factor-start suffix is strictly smaller than every earlier factor-start suffix; repeated equal factors are included in the proper-prefix case of the reduction above.

Now take a position q strictly inside u_h , and write v for the proper suffix of u_h beginning there. Since u_h is Lyndon, $u_h < v$. The two words are not in a prefix relation: v is shorter, while v cannot be a prefix of u_h because that would make it a border. Appending the same later factor tail therefore preserves the strict comparison, so S_h is smaller than the suffix beginning at q .

If $i = s_j$ is a factor start, every earlier position is either an earlier factor start, whose suffix is larger by the first paragraph, or lies inside an earlier factor, whose suffix is larger than that factor-start suffix by the second paragraph. Hence (4) holds. Conversely, if i lies inside u_h , the earlier position s_h has a strictly smaller suffix, so i is not a new minimum. This proves both directions. $\square$

Remark 2.2. Strictness matters when consecutive factors are equal. For example, the CFL factorization of 00 is 00 , and the second suffix 0 is strictly smaller than the first suffix 00 because it is a proper prefix.

3. THE LEADING-ONE AMPLIFIER

Lemma 3.1 (literal prefix rule). *If $w = 1^r 0s$ with $0 \leq r < n$, then*

$$(5) \quad L_n(w) = 1^r L_{n-r}(0s).$$

More generally, $L_{m+1}(1v) = 1L_m(v)$ for every $v \in \{0, 1\}^m$.

Proof. A binary Lyndon word of length greater than one begins with zero: a word beginning in one and containing a zero has a smaller rotation beginning at that zero, while 1^d is not primitive for $d > 1$. Hence every leading one in w is a singleton Lyndon factor. The first factor of the nonempty remainder $0s$ begins in zero and is therefore smaller than the singleton 1. Uniqueness of (1) shows that the full factorization is the r singleton factors followed by the CFL factorization of $0s$, proving (5).

For arbitrary v , either $v = 1^m$, when the last statement is immediate, or $v = 1^q 0s$ and (5) applied with $r = q$ and $r = q + 1$ gives the identity. $\square$

Let $\lambda(w)$ be the number of leading ones in w .

Theorem 3.2 (recurrent atlas and clock bound). *The word 1^n is the unique fixed point and the unique recurrent state of L_n . Every orbit reaches it in at most n updates.*

Proof. The CFL factorization of 1^n consists of n singleton factors, so 1^n is fixed. If $w \neq 1^n$, write $w = 1^r 0s$. Equation (5) shows that the first r output bits are one, and the first bit of $L_{n-r}(0s)$ is also one because every nonempty factorization has a first factor. Thus $\lambda(L_n(w)) \geq r + 1$. The integer λ strictly increases at every nonfixed update and is bounded by n . No other recurrent state can exist, and at most n updates are required. $\square$

4. THE SHARP AND UNIQUE DEEPEST WORD

Let $a_n = 0101 \cdots$ be the length- n alternating word beginning in zero.

Lemma 4.1 (alternating recursion). *For every $n \geq 1$,*

$$(6) \quad L_n(a_n) = 1a_{n-1},$$

where a_0 is empty.

Proof. The word 01 is Lyndon. If $n = 2m$, the CFL factorization of a_n is $(01)^m$, written as m equal factors. If $n = 2m + 1$, it is $(01)^m 0$; this is nonincreasing because $01 > 0$. The starts are exactly the odd positions, which is the mask $1a_{n-1}$. At $n = 1$, the factorization is the singleton 0 and the same identity holds. $\square$

Theorem 4.2 (sharp clock and uniqueness). *The maximum depth of L_n is n . Its unique depth- n state is a_n .*

Proof. By lemma 3.1, prefixing a word by one does not change its depth: the identity $L_{m+1}(1v) = 1L_m(v)$ persists under iteration, including when v becomes all ones. Therefore (6) and induction give

$$\text{depth}(a_n) = 1 + \text{depth}(a_{n-1}) = n,$$

with $a_1 = 0 \mapsto 1$. This attains the upper bound in theorem 3.2.

We prove uniqueness by induction. At $n = 1$, the word 0 is the only state of depth one. Let $n \geq 2$ and suppose w has depth n . Its image has depth $n - 1$ and begins in one. Write $L_n(w) = 1v$. Prefix invariance of depth and the induction hypothesis force $v = a_{n-1}$, so

$$(7) \quad L_n(w) = 1a_{n-1}.$$

The one-positions in this mask prescribe CFL factor lengths $(2, 2, \dots, 2)$ when n is even and $(2, 2, \dots, 2, 1)$ when n is odd. The only binary Lyndon word of length two is 01. In the odd case, the last singleton cannot be 1, because the required nonincrease would read $01 \geq 1$, which is false; it must be 0. Hence $w = (01)^{n/2}$ for even n , and $w = (01)^{(n-1)/2}0$ for odd n . In both cases $w = a_n$. $\square$

5. EVERY TARGET FIBRE AS AN ORDERED LYNDON CHAIN

For $d \geq 1$, let $\mathcal{L}_d$ be the lexicographically ordered set of binary Lyndon words of length d . If a target $y \in \{0, 1\}^n$ begins in one, its one-positions determine a composition $\ell(y) = (\ell_1, \dots, \ell_k)$ of n : these are the successive gaps between factor starts, with the last gap ending at $n + 1$.

Theorem 5.1 (complete inverse atlas). *If $y_1 = 0$, then $L_n^{-1}(y)$ is empty. If $y_1 = 1$ and $\ell(y) = (\ell_1, \dots, \ell_k)$, then*

$$(8) \quad |L_n^{-1}(y)| = \#\{(u_1, \dots, u_k) : u_i \in \mathcal{L}_{\ell_i}, u_1 \geq \dots \geq u_k\}.$$

Thus positivity of the right-hand side is the complete image criterion.

Proof. Every factor-start mask begins with one, giving the first assertion. For a target beginning in one, any preimage has factors starting at exactly its one-positions, so their lengths are $\ell_1, \dots, \ell_k$. By the CFL theorem, those factors are Lyndon and nonincreasing, which maps the fibre into the set on the right of (8). Conversely, concatenate any tuple on the right. It is already a nonincreasing factorization into Lyndon words; uniqueness of (1) makes it the CFL factorization of the concatenation, whose start mask is y . The two constructions are inverse. $\square$

The count has a finite matrix form. For positive a, b , let $M_{a,b}$ be the $|\mathcal{L}_a| \times |\mathcal{L}_b|$ zero-one matrix

$$(9) \quad M_{a,b}(u, v) = \mathbf{1}\{u \geq v\}, \quad u \in \mathcal{L}_a, v \in \mathcal{L}_b,$$

and let $\mathbf{1}_d$ be the all-one column indexed by $\mathcal{L}_d$.

Corollary 5.2 (matrix fibres and two closed cells). *For $k \geq 2$, the fibre in (8) equals*

$$(10) \quad \mathbf{1}_{\ell_1}^T M_{\ell_1, \ell_2} \cdots M_{\ell_{k-1}, \ell_k} \mathbf{1}_{\ell_k};$$

for $k = 1$ it equals $|\mathcal{L}_{\ell_1}|$. Moreover,

$$(11) \quad |L_n^{-1}(10^{n-1})| = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d},$$

$$(12) \quad |L_n^{-1}(1^n)| = n + 1.$$

Proof. Expanding (10) sums one for each tuple satisfying all adjacent inequalities, exactly as in (8). The target 10^{n-1} prescribes the one-part composition (n) , so its fibre is $\mathcal{L}_n$. The classical primitive-necklace Möbius formula gives (11). For 1^n , every factor has length one. The two one-letter Lyndon words are 0 and 1, and a nonincreasing sequence is uniquely $1^j 0^{n-j}$ for some $0 \leq j \leq n$, proving (12). $\square$

6. EXACT CONTROL AND LIMITATIONS

The paper-local verifier independently computes the Duval mask and the strict suffix-record mask for every binary word through $n = 18$. It constructs all functional graphs in that range, totaling 524,286 states, and checks the fixed atlas, clock, and unique deepest word. A separate Lyndon-word generator and rectangular-matrix evaluator agrees with every literal target fibre through $n = 14$; the two special fibres are checked through $n = 18$. The canonical run executes 2,654,300 exact assertions.

Finite enumeration is only counterexample pressure. The observed image and largest-fibre sequences are not promoted to unproved all-length statements. The owner search is bounded, so a non-hit is not a novelty or priority certificate. This anonymous artifact remains `HOLD_EXTERNAL`; public posting, submission, and release are not authorized.

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